Flight Test and Evaluation Report

Aircraft: MiG-29UB (Fulcrum-B) · Test subject: Steady level flight · Document status: Illustrative technical report



Aircraft and System Testing Establishment

Flight Test and Evaluation Report No.: FTER/ILL/029/001

Report title: Evaluation of Level-Flight Characteristics of the MiG-29UB Aircraft

Report classification: Illustrative training document — not an approved flight-test instruction or operational publication

Aircraft variant

MiG-29UB, twin-seat trainer

Evaluation area

Steady, unaccelerated level flight

Prepared by

[Test Engineer / Author]

Date

[Insert date]

Document control

Field	Details
Report number	FTER/ILL/029/001
Revision	0
Prepared by	[Name / Appointment]
Reviewed by	[Flight Test Engineer]
Approved by	[Authorised Approving Authority]
Reference records	To be populated from approved aircraft documentation
Evidence status	No actual flight-test data supplied

1. Executive summary

This report establishes a technical framework for evaluating the level-flight characteristics of the MiG-29UB twin-seat trainer. The assessment considers the physical principles governing steady level flight, the relationship between aircraft weight and aerodynamic lift, the significance of airspeed and altitude measurements, engine-performance indications, aircraft handling observations, instrumentation quality, and the assessment of measurement uncertainty.

The MiG-29UB is a two-seat training variant of the MiG-29 family. Published descriptions identify its twin-engine configuration and its role in pilot conversion and training. The variant differs from the single-seat aircraft in configuration and some equipment, so performance figures for another variant cannot automatically be treated as MiG-29UB test results.

The principal evaluation objective is to establish whether recorded flight data demonstrate stable altitude, consistent airspeed, acceptable aircraft handling, and repeatable performance under documented conditions.

Result status: No flight has been documented in the information supplied. Consequently, no pass/fail verdict, measured performance value, or airworthiness conclusion can be issued. The report below provides the technical analysis framework and the fields required for a properly evidenced evaluation.

2. Introduction

2.1 Background

Level flight is a fundamental aircraft-performance condition in which the aircraft maintains approximately constant altitude and constant velocity relative to the surrounding air. It is used as a basis for examining aerodynamic balance, engine performance, stability, handling qualities, and the consistency of recorded flight data.

For a twin-seat military training aircraft, evaluation of this condition is also relevant to understanding instrument indications and documenting aircraft behaviour during routine flight.

2.2 Aim

To establish a structured method for evaluating the level-flight characteristics of the MiG-29UB using verified aircraft records, authorised flight-test data, and established aerodynamic principles.

2.3 Objectives

Establish the aircraft configuration and test-data provenance.

Define the meaning of steady, unaccelerated level flight.

Identify the principal parameters relevant to performance evaluation.

Explain the theoretical relationships between lift, drag, thrust, and weight.

Define the framework for assessing data quality and measurement uncertainty.

Record limitations, observations, and evidence-based conclusions.

2.4 Scope

The report covers:

Steady level-flight aerodynamics.

General performance-parameter definitions.

Instrumentation and data-quality considerations.

Data reduction and uncertainty assessment.

Reporting of observations and limitations.

It does not provide a flight-test card, aircraft-specific operating settings, manoeuvre instructions, performance-envelope exploration, or instructions for operating the aircraft. Such activities require approved technical documentation and appropriately authorised personnel.

3. Aircraft description





3.1 General characteristics

The following table summarises publicly described characteristics. These are identification-level reference details, not measurements obtained during the evaluation described in this report.

Parameter	General description
Aircraft	MiG-29UB
NATO reporting name	Fulcrum-B
Aircraft type	Twin-seat jet trainer
Crew arrangement	Two seats in tandem
Powerplant	Two Klimov RD-33 turbofan engines
Wing arrangement	Mid-wing configuration
Tail arrangement	Twin vertical stabilisers
Primary training role	Pilot conversion and operational training

Source: Published MiG-29UB reference material.

3.2 Configuration considerations

The actual aircraft configuration must be documented before interpreting flight data. Relevant record fields include aircraft identity, modification status, mass properties, centre-of-gravity documentation, external configuration, instrumentation fit, and maintenance status.

These details can materially influence the interpretation of aerodynamic and engine-performance measurements. Values must be taken from the specific aircraft's approved records rather than inferred from general descriptions of the MiG-29 family.

4. Test objectives and evaluation criteria

The evaluation is organised around the following evidence requirements.

Evaluation	Evidence required	Assessment basis
Flight	Recorded altitude and velocity history	Consistency with the documented
Aerodynamic balance	Validated flight parameters and configuration records	Agreement with the governing aerodynamic relationships
Engine performance	Recorded engine indications and applicable approved references	Comparison against authorised limits and reference data
Handling observations	Recorded observations from authorised personnel	Approved handling-quality criteria
Instrumentatio	Calibration records and data-quality checks	Measurement validity and traceability
Repeatability	Comparable, valid data sets	Agreement within established
Overall	Complete evidence and applicable	Formal review by authorised

Acceptance limits must be specified by the responsible test authority before a real evaluation. They cannot be derived merely from public maximum-performance figures.

5. Conditions relevant to the evaluation

5.1 Aircraft condition

The report should identify the aircraft and its configuration at the time of the recorded evaluation. Relevant evidence includes:

Aircraft identity and variant.

Applicable modification and configuration records.

Approved mass and centre-of-gravity records.

Relevant maintenance and instrument-calibration status.

Recorded equipment configuration.

Any documented defect, limitation, or data-quality issue.

5.2 Environmental conditions

Environmental information is necessary to interpret aerodynamic measurements and compare results obtained under different conditions.

The data record should identify the source and validity of the following information:

Parameter	Why it matters
Atmospheric pressure	Supports pressure-altitude and air-data interpretation
Ambient temperature	Affects air density and engine-performance interpretation
Wind conditions	Relevant to ground-referenced velocity and position
Atmospheric density	Influences lift, drag, and air-data relationships
Atmospheric variability	Can affect the comparability of measurements

Missing or unreliable environmental data should be documented as a limitation rather than replaced with assumed values.

5.3 Test personnel and recording arrangements

The completed report should identify the responsible test personnel, their roles, the source of each data set, the recording system used, and the applicable authorisation records.

No personnel names, dates, weather conditions, or test events should be represented as actual when they have not been documented.

6. Theoretical basis of level flight

6.1 Definition

For idealised, steady, unaccelerated flight in a straight line, the principal force-balance relationships are:

$$ L=W $$
$$ T=D $$
where:

L = aerodynamic lift

W = aircraft weight

T = thrust along the flight direction

D = aerodynamic drag

These relationships assume that the aircraft is in steady flight and that the relevant force components have been resolved along the appropriate axes. More general analyses must account for flight-path angle, acceleration, and other force components.

6.2 Lift relationship

The general aerodynamic lift equation is:

$$ L=\frac{1}{2}\rho V^2 S C_L $$
where:

ρ = local air density

V = true airspeed

S = reference wing area

C_L = lift coefficient

The equation shows that lift depends on air density, airspeed, reference area, and the lift coefficient. The lift coefficient itself depends on the aerodynamic state and configuration.

In steady level flight, lift balances weight. Therefore, changes in weight, density, or airspeed are relevant when interpreting the aerodynamic condition. This relationship alone does not establish the actual lift coefficient or performance of a specific aircraft.

6.3 Drag and engine performance

Aerodynamic drag is commonly represented as the sum of parasite and induced contributions in a simplified model:

$$ C_D=C_{D0}+kC_L^2 $$
where C_{D0} represents the zero-lift drag coefficient in the model and k is an induced-drag factor.

This simplified relationship is useful for explaining aerodynamic trends, but its coefficients must be established from appropriate data before it can be used for quantitative aircraft-specific analysis.

In steady level flight, the thrust required to balance drag is related to the aircraft's aerodynamic condition. Engine indications alone do not directly measure aerodynamic drag, and a quantitative assessment requires suitable validated data.

6.4 Power required

Aerodynamic power required can be expressed as:

$$ P_{\mathrm{required}}=D V $$
This relationship describes the rate at which work is required to overcome aerodynamic drag. It is not, by itself, a measure of engine fuel consumption or installed engine efficiency.

6.5 Interpretation of the theoretical model

The equations in this section establish the scientific basis for analysing level flight. They do not constitute a verified performance model of the MiG-29UB. Aircraft-specific conclusions require validated coefficients, approved configuration data, reliable measurements, and a documented uncertainty assessment.

7. Instrumentation and data quality

7.1 General data categories

A formal evaluation may require appropriately authorised and validated records covering:

Data category	Intended analytical use
Air-data records	Establishing the recorded airspeed and altitude condition
Aircraft attitude records	Assessing the recorded flight state
Position and time records	Establishing traceability and temporal consistency
Engine-recording data	Assessing recorded engine behaviour
Mass-property records	Establishing the aircraft's documented loading condition
Environmental records	Supporting interpretation of atmospheric effects
Calibration documentation	Establishing measurement validity

his list identifies evidence categories, not a prescribed aircraft-specific instrumentation installation.

7.2 Calibration and traceability

Each quantitative result should be traceable to its original measurement source. The completed report should record, where applicable:

Instrument identification and calibration status.

Measurement units and conversion methods.

Data-acquisition timestamps.

Missing, invalid, or rejected records.

Processing software and calculation versions.

Rounding and significant-figure conventions.

The person responsible for data validation.

7.3 Data acceptance

A data set should not be treated as valid solely because it contains plausible-looking numbers. The reviewer should verify its provenance, units, completeness, consistency, and conformity with the applicable data-quality criteria.

8. Data reduction and analysis

This section defines how results should be presented once verified data are available. It intentionally does not supply aircraft-specific flight settings or fabricated measurements.

8.1 Data-record template

Parameter	Unit	Recorded value	Uncertainty	Source
Aircraft identifier	—	[Enter]	N/A	Aircraft records
Configuration reference	—	[Enter]	N/A	Configuration record
Mass reference	kg	[Enter]	[Enter]	Approved mass
Pressure altitude	m or ft	[Enter]	[Enter]	Validated air-data record
Airspeed type	—	[IAS/CAS/TAS as recorded]	N/A	Instrument definition
Airspeed value	Applicable unit	[Enter]	[Enter]	Validated air-data record
Ambient temperature	°C or K	[Enter]	[Enter]	Environmental record
Engine data	Applicable unit	[Enter]	[Enter]	Validated engine record
Data validity	—	[Valid/Invalid/Review]	N/A	Data-quality review

The table is a blank recording template, not a set of measured results.

8.2 Consistency assessment

The reviewer should determine whether the recorded altitude and velocity histories support the claimed steady level-flight condition, taking account of measurement resolution and the pre-established acceptance criteria.

Any departures from the intended condition should be recorded and considered when deciding whether a data set is suitable for analysis.

8.3 Comparison with reference data

A valid comparison requires the test configuration and the reference configuration to be sufficiently comparable. Differences in variant, mass, external configuration, atmospheric conditions, instrument definitions, and data-processing methods may make a direct comparison invalid.

Publicly reported maximum speed, range, or service ceiling must not be substituted for measurements of steady level-flight performance. Public sources themselves can differ in definitions and configuration assumptions.

9. Results and observations

9.1 Results status

No primary flight-test data were supplied for this report. Accordingly, the following fields remain unpopulated.

Result item	Status
Verified level-flight data	Not supplied
Measured airspeed and altitude stability	Not evaluated
Measured engine performance	Not evaluated
Instrumentation uncertainty	Not calculated
Repeatability	Not evaluated
Compliance with acceptance criteria	Undetermined

9.2 Observation register

Observation ID	Observation	Evidence reference	Disposition
OBS-01	Flight-test data are unavailable in the	Not	Obtain verified data
OBS-02	Aircraft-specific acceptance criteria have not been supplied	Not applicable	Confirm applicable approved criteria
OBS-03	Measurement uncertainty has not been	Not	Complete data-quality

These are documentation observations, not findings of aircraft defects.

9.3 Graphs to include in a completed report

The following plots may be useful when suitable verified data exist:

Recorded altitude against time.

Recorded airspeed against time.

Recorded engine parameters against time.

Comparison of repeat measurements with uncertainty ranges.

No plots are populated here because doing so would require actual data or clearly identified simulated data.

10. Measurement uncertainty

All quantitative evaluations should consider the uncertainty associated with the measurements and subsequent calculations.

For a derived result $y=f(x_1,x_2,\ldots,x_n)$, a first-order approximation for combined standard uncertainty is:

$$ u_y^2\approx \sum_{i=1}^{n} \left(\frac{\partial f}{\partial x_i}\right)^2u_{x_i}^2 $$

This form assumes the input quantities are uncorrelated. If they are correlated, covariance terms must also be included.

A completed uncertainty assessment should identify the measurement sources, calibration uncertainties, data-processing contributions, relevant correlations, and the method used to calculate the reported uncertainty.

No numerical uncertainty is reported in this document because the necessary input measurements and calibration records are unavailable.

11. Limitations

The following limitations apply:

The report is a technical template rather than a record of a completed flight.

No aircraft-specific flight-test data have been provided.

Publicly available descriptive specifications are not equivalent to controlled engineering data.

No aircraft-specific performance coefficients have been established here.

No formal compliance determination or airworthiness assessment can be made.

The report does not replace the aircraft's approved flight manual, maintenance documentation, or authorised test procedures.

12. Conclusion

The report establishes a structured basis for evaluating the MiG-29UB in steady level flight. The theoretical analysis confirms that the principal idealised relationships involve lift balancing weight and thrust balancing drag, subject to the assumptions of steady, unaccelerated flight.

A defensible aircraft-specific conclusion requires validated flight-test records, a documented configuration, appropriate environmental information, established acceptance criteria, and an assessment of measurement uncertainty.

Final evaluation status: INDETERMINATE — awaiting verified test evidence.

This status means that the available information is insufficient to reach a performance verdict. It does not indicate either satisfactory or unsatisfactory aircraft performance.

13. Recommendations

Populate the report using authorised, traceable aircraft and flight-test records.

Verify the aircraft variant and configuration before comparing measurements with reference data.

Document all acceptance criteria and measurement uncertainties.

Ensure that all calculated results can be reproduced from the recorded source data.

Obtain the required technical review and formal approval before designating the document as an official flight-test report.

Appendix A — Report completion checklist

Evidence completion

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Aircraft identity and configuration verified
Authorised test records attached
Environmental conditions documented
Measurement sources and calibration status verified
Data quality reviewed
Calculations independently checked
Uncertainty assessment completed
Observations and limitations documented
Technical review completed
Required approval recorded
This checklist tracks document preparation only. Checking an item does not independently validate evidence or confer official approval.

